

2025 Pennsylvania Apple and Peach Phenology Report - April 14

The apple and peach phenology report tracks bud development from the start of the season through the setting of fruit in four growing regions in Pennsylvania: Central, Northwestern, Southcentral, and Southeastern.

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Updated: April 15, 2025



Gala at tight cluster in Centre County, PA. (Photo credit: Lindsay Brown, Penn State University.)

Phenology Report and Damage Update

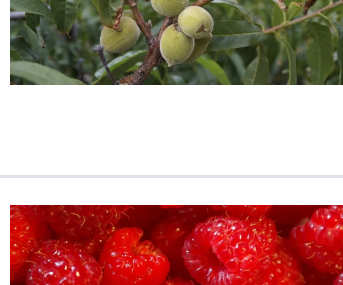
Freezing conditions were observed throughout Pennsylvania during the week of April 6-12, 2025. Due to the extent of this damage, a separate report documenting observed freeze damage in apples and peaches will be available soon.

This report will focus on the phenology update for the week and readers are urged to consider both reports worth reading.

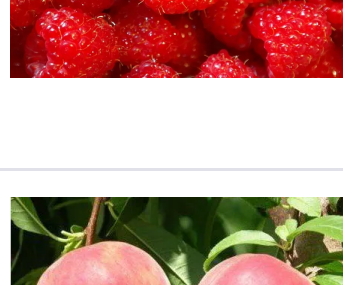
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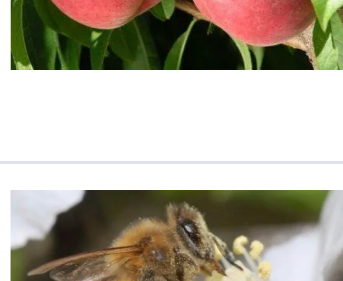
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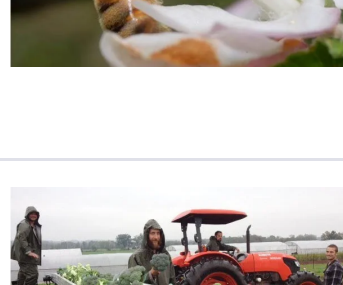
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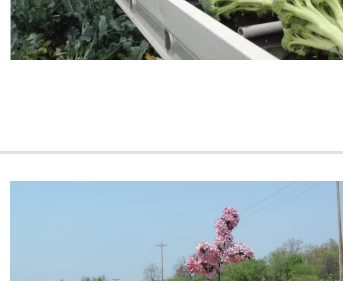
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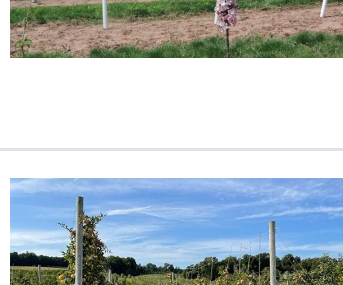
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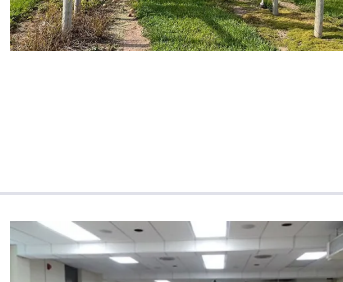
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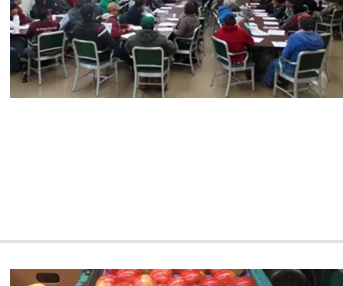
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Phenology Data

Growing Degree Day Accumulation

Growing Degree Day (GDD) data taken from NEWA using the Degree Day Type of "Base 43 °F" and "Base 50 °F" between the dates of 1/1/2025 and 4/14/2025.

County	NEWA Weather Station	GDD43 °F	GDD50 °F
Adams	Biglerville (FREC)	288	110
Berks	Boyetown	241	79
Centre	Rock Spring (Penn State Orchard)	158	52
Erie	Harborcreek	171	59
Franklin	Orrtanna*	307	123

*Growing Degree Day (GDD) data taken from [NEWA](#)

Apple Bloom Phenology

This chart documents the standard bud stage names for apple blossom development and lists the temperatures at which there is 10% and 90% bud kill after a 30-minute exposure to the given temperatures. Use it to interpret the bud stages in the tables below, and to monitor potential damage in your apple orchards when frost or freeze events are expected.

A P P L E								
	Silver Tip	Green Tip	Half-Inch Green	Tight Cluster	First Pink (Pink)	Full Pink (Open Cluster)	First Bloom (King Bloom)	Full Bloom and Post-bloom
	10% 15 2	18 23 10	23 28 15	27 32 21	28 33 24	28 33 25	28 33 25	28 33 25

Figure from the Utah State Extension fact sheet "[Critical Temperatures for Frost Damage on Fruit Trees](#)."

Temperatures are in °F. Images used by permission of the author, Marion Murray, Utah State University.

Phenology estimations based on observations made on the morning of April 14th, 2025 for

Apple Varieties

County	Fuji	Gala	Golden Delicious	Honeycrisp	Pink Lady®	Red Delicious
Adams	First Pink	First Pink	First Pink	First Pink	First Pink	First Pink
Berks	Tight Cluster	Tight Cluster	Tight Cluster	Tight Cluster	Pink	Pink
Centre	Tight Cluster	Tight Cluster	Tight Cluster	Tight Cluster	Tight Cluster	Tight Cluster
Erie	Green Tip	Green Tip	Green Tip	Green Tip	Green Tip	Green Tip
Lawrence	Tight Cluster	Tight Cluster	Tight Cluster	Tight Cluster	Pink	First Pink

Peach Bloom Phenology

This chart documents the standard bud stage names for peach blossom development and lists the temperatures at which there is 10% and 90% bud kill after a 30-minute exposure to the given temperatures. Use it to interpret the bud stages in the tables below, and to monitor potential damage in your peach orchards when frost or freeze events are expected.

P E A C H							
	Swollen Bud (First Swell)	Calyx Green	Quarter-Inch Green (Calyx Red)	Pink (First Pink)	First Bloom	Full Bloom	Post-bloom
	10% 18 1	21 26 5	23 28 9	25 30 15	26 31 21	26 31 24	28 33 25

Figure from the Utah State Extension fact sheet "[Critical Temperatures for Frost Damage on Fruit Trees](#)."

Temperatures are in °F. Images used by permission of the author, Marion Murray, Utah State University.

Varieties observed per county (Early, Mid-, and Late Season)

County	Early Variety	Early Stage	Mid Variety	Mid Stage	Late Variety	Late Stage
Adams	Desiree®	Full Bloom	John Boy®	Full Bloom	Coralstar®	50% First Bloom
	Sentry	Full Bloom	--	--	Messina®	50% First Bloom
	SilverGlo™	Full Bloom	--	--	--	--
Berks	Desiree®	First Bloom	Redhaven	First Bloom	Messina®	Full Bloom
Centre	--	--	Redhaven	Calyx Red	--	--
Franklin	Desiree®	72% Pink / 25% Full Bloom / 2% Post Bloom	John Boy®	10% Pink / 85% Full Bloom / 5% Post Bloom	Bounty	35% Pink / 60% Full Bloom / 5% Post Bloom
	Glenglo™	40% Pink / 60% Full Bloom	Redhaven	20% Pink / 80% Full Bloom	Messina®	98% Full Bloom / 2% Post Bloom
	--	--	--	--	--	--
Lawrence	<various>	Pink	Redhaven	Pink	Coralstar®	Pink
	--	--	--	--	Autumnstar®	Pink

Additional Critical Temperature Charts

For more details on critical temperatures in other tree fruit crops, please download the chart from the [Utah State Extension](#) website. Please keep in mind that significant temperature fluctuations could be damaging to fruit buds.

If you plan on using wind machines for damage mitigation, review this resource on [temperature inversion potentials](#) and the correct conditions for orchard fan use.

For more information on frost mitigation techniques and background information, this [article on frost, critical temperatures, and frost protection](#) can help.

Special Note on Blossom Thinning of Peaches with Accede™

While the risk of frost is low this week throughout much of Pennsylvania, please monitor weather conditions closely. To estimate damage, cut through the buds longitudinally and check for browning or blackening of the pistils or ovules, which indicates tissue death. Avoid placing the shoots in water to observe bloom, as damaged buds may still bloom but often fail to set fruit due to damaged reproductive organs.

Once you've assessed the extent of damage from the freeze event, you can determine the appropriate application rate for Accede™. The labeled use rate is 300–600 ppm, which is equivalent to 10–20 ounces of soluble granule product in 100 gallons of water/acre. Adjust the rate as needed based on the severity of the damage. As a general use rate, 15 oz. (450 ppm) of Accede™ in 100 gallons of water/acre is recommended. Please reduce rates if the bloom is below normal, reduce rates for young trees, sensitive varieties, as well frost damaged trees.

References

Rob Crassweller. Orchard Frost - Protection with Wind Machines. Penn State Extension. [Orchard Frost - Protection with Wind Machines](#)

Rob Crassweller. Frost, Critical Temperatures, and Frost Protection. Penn State Extension. [Frost, Critical Temperatures, and Frost Protection](#)

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